Give Your .NET Agent a Face

Building an AI desk Companion with M5Stack



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Give an agent a face, a voice, and movement

Meet StackChan

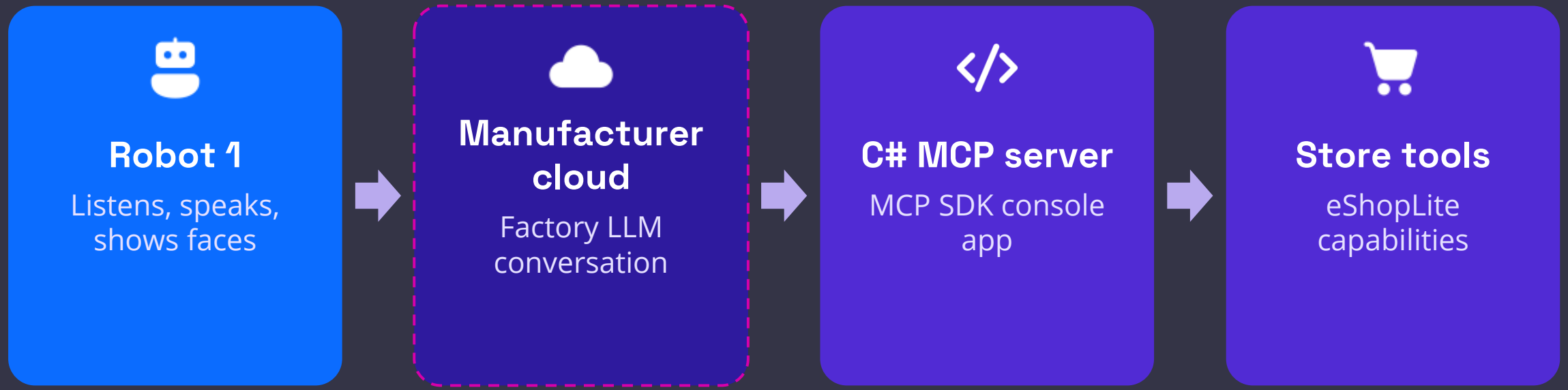
From M5Stack

- Listens
- Speaks
- Shows faces

A desktop robot with an expressive screen for a face



From conversation to tools



Conceptual roles. Dashed: manufacturer service, not ours. Connection details to be verified

StackChan + Your Tools

Conceptual architecture

Factory firmware, cloud conversation, custom MCP tools



StackChan

Factory firmware

Wi-Fi + Internet

Conversation

M5Stack / Xiaozhi cloud

Conversation service

Handles the robot conversation and context

Coordinates tool use

MCP endpoint

Connects external MCP tools to the conversation service

Connect with token

Outbound secure WebSocket

MCP tool calls + results

Your tools server

.NET MCP server

MCP connector

Establishes the outbound cloud connection

Routes MCP requests and results

Your custom tools

Example tools

Search **Query** **Actions**

The cloud invokes your tools during the robot conversation

A C# connection check

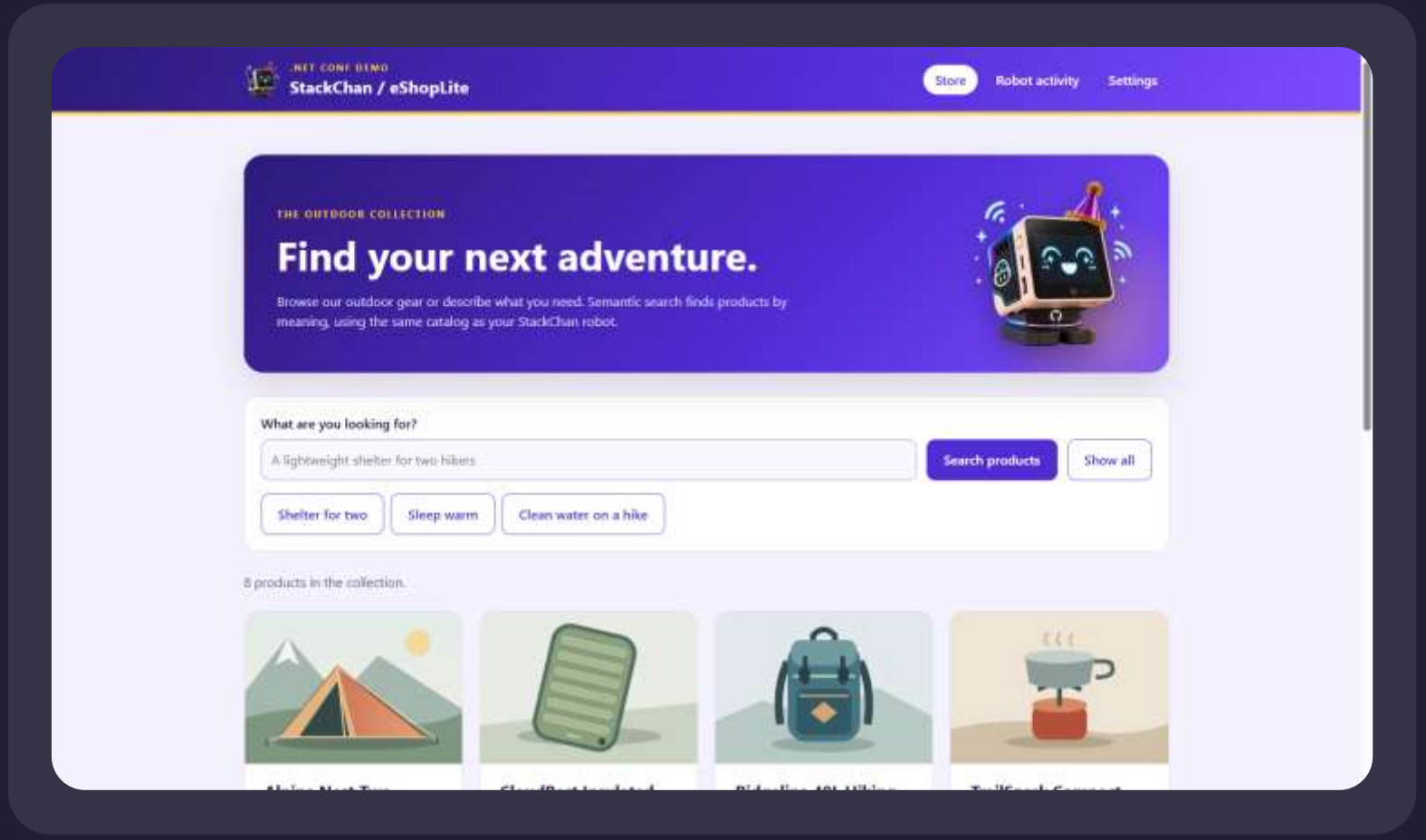
```
// Remote connection check (diagnostic, not the eShopLite tool bridge)
// Endpoint loaded from user secrets or environment variables

.AddUserSecrets<ConnectionProbe>(optional: true)
.AddEnvironmentVariables().Build()
ConnectionOptions.GetRequiredEndpoint(configuration)
using var socket = new ClientWebSocket();
await socket.ConnectAsync(endpoint, deadline.Token);

// eShopLite tool excerpt: follow-up from the real tool source
```

Meet eShopLite

- Outdoor gear for camping, mountains, and climbing
- Semantic search, prepared English example
- Runs locally with Aspire



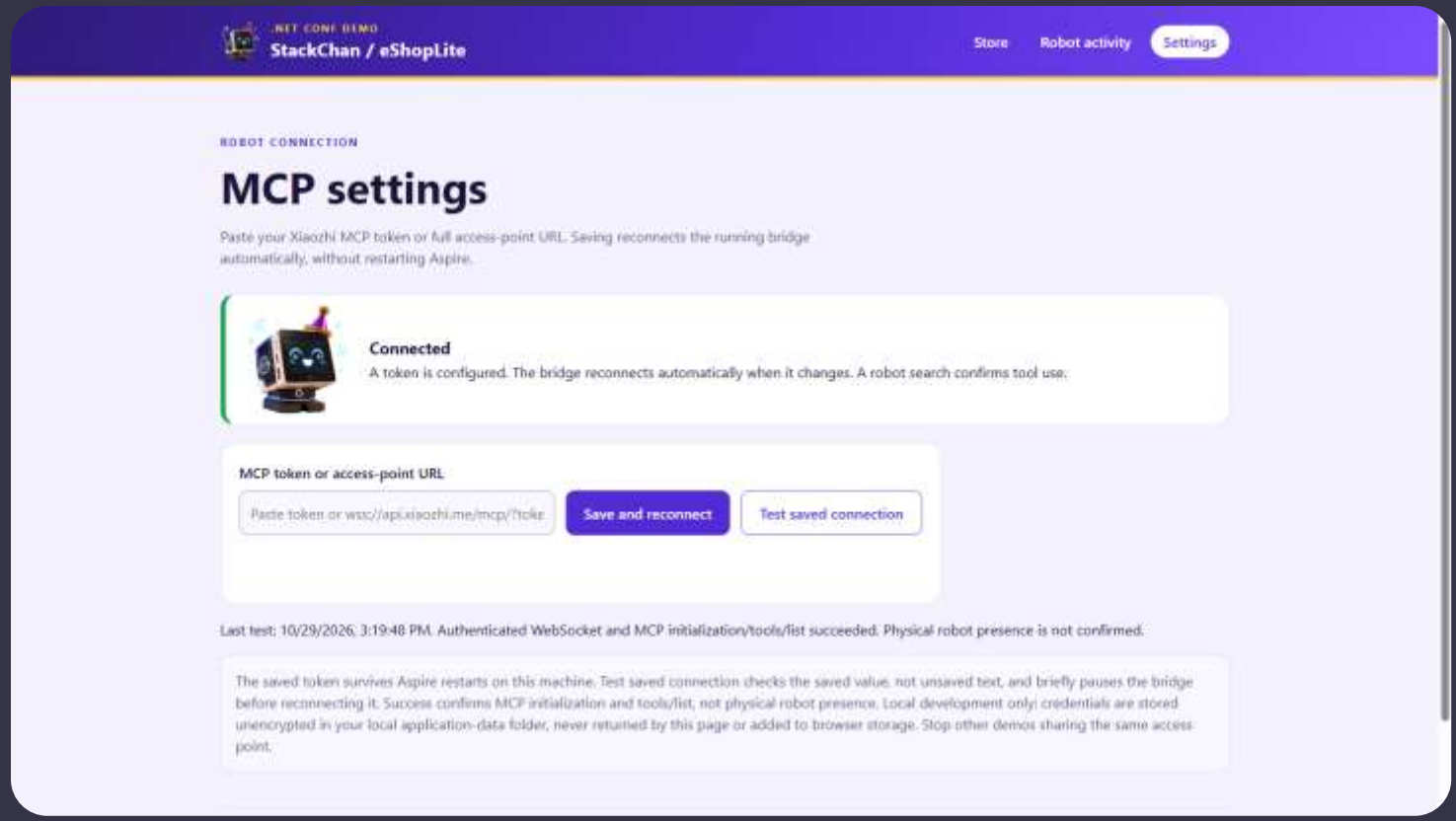
Connect once, demonstrate safely

Token-based configuration

Connection prepared before the session

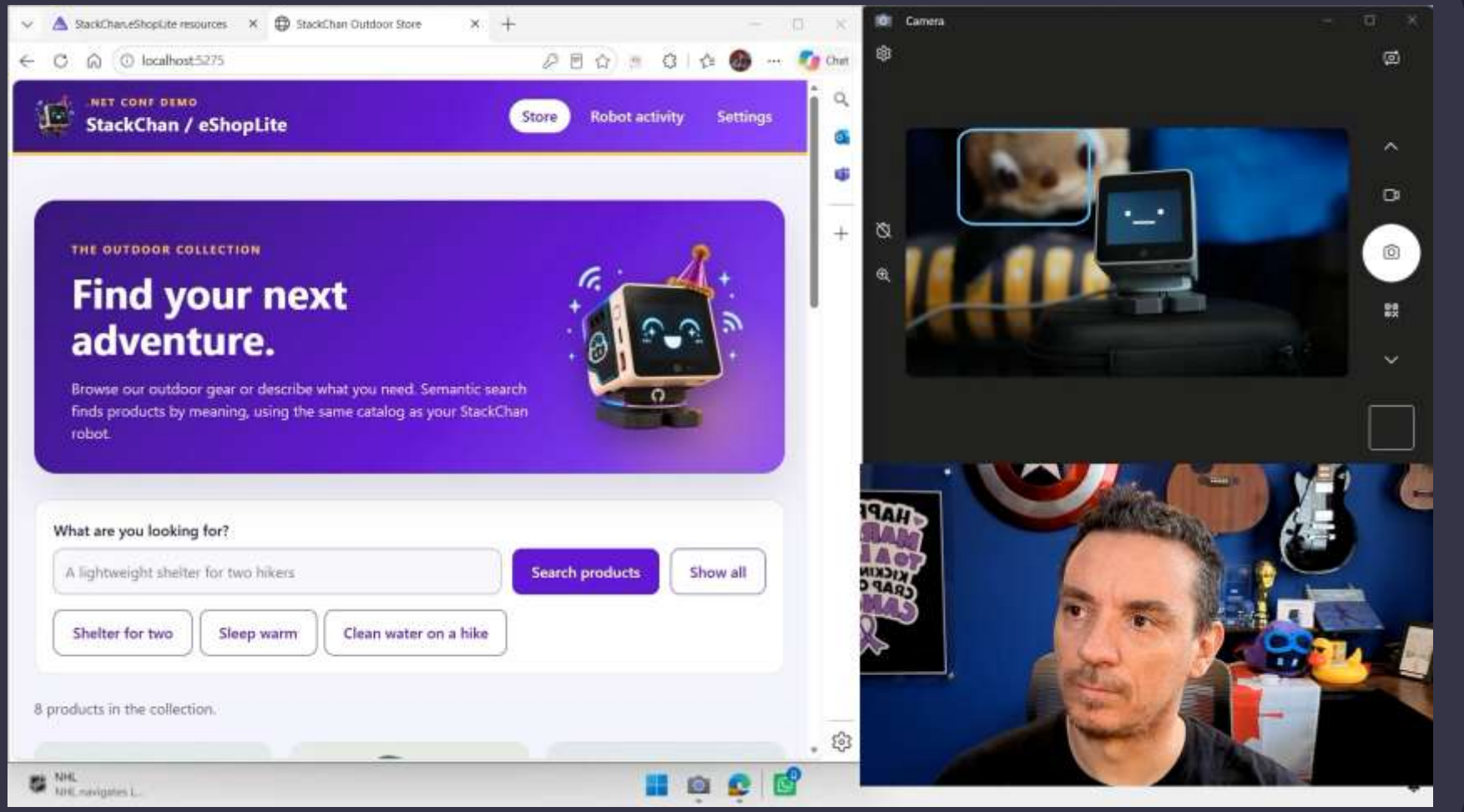
Token stays private and never appears on screen

No live configuration



Static screenshot: connection settings, token field empty

eShopLite meets StackChan



New Year, New Bot

A cute desk companion with a .NET brain

Proposed architecture · Desktop phase

aka.ms/newyearnewbot



StackChan

Custom C++ firmware

Wi-Fi

HTTP on trusted LAN

.NET backend

ASP.NET Core · .NET 11

RobotHub

Robot behavior and notification policy

Robot controls and companion API

Notification candidates

Work agent

Disabled by default

Microsoft Agent Framework

Foundry path: opt-in, synthetic data only

GitHub Copilot SDK · Planned

Aspire · Orchestration · Logs · Traces · Metrics

External integrations

AI requests

Responses

Microsoft Foundry

AI models

Opt-in · Synthetic demo

WorkIQ MCP · Planned

Read-only work context

Microsoft 365

Email + Teams

Planned

Implemented components

Planned integrations

Opt-in badge: explicitly enabled synthetic demo path

Stream mode: generic robot alerts · Private work details stay in the dashboard

Optional: a canvas for the workflow

Robot Delivery Monitor: firmware workflow in five steps



Conceptual illustration, not a screenshot or an agent run

A different robot, our own firmware

Robot 2, not the factory device

- Custom C/C++ firmware built from scratch
- Animated .NET avatar
- Private Wi-Fi
- Status, connectivity, and firmware version



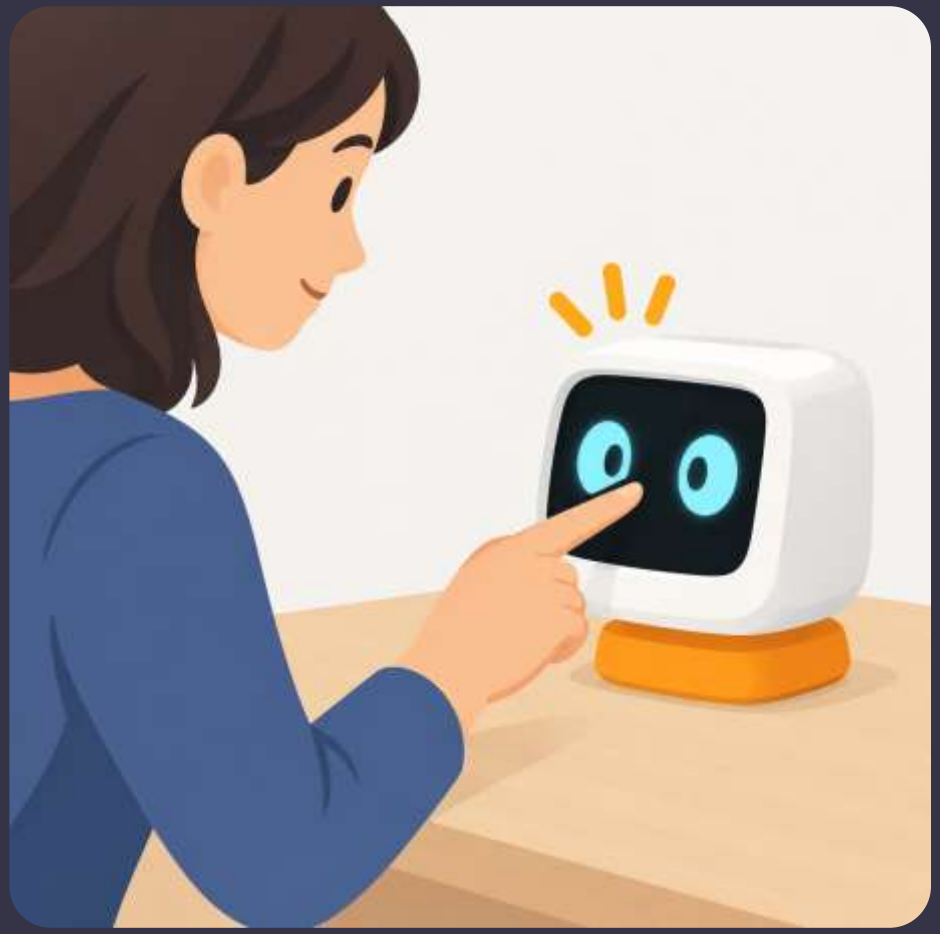
Send a notification, see it arrive on the desk

Live demo: a notification on the desk

1 Robot 2 camera

2 Sanitized control page

3 Notification arrives on the desk



Give your agent a face

Join New Year, New Bot

aka.ms/newyearnewbot



